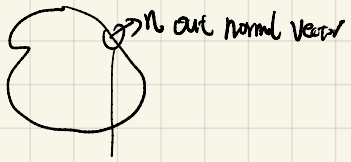
Above design is general for all DG, Now we focus on Euler!

Here, everything is not dependent on init-condition or specific problem

Prefer Shu. Section 2



$$d\vec{F}_p = -p\,n\,ds$$

$$\vec{F}_p = -\int_{\partial\Omega} p\,n\,ds$$

Newton-second

$$\frac{d}{dt}\int_\Omega \rho\vec{u} = -\int_{\partial\Omega} \rho\vec{u}(\vec{u}\cdot n)\,ds - \int_{\partial\Omega} p n\,ds$$

$$\rho u \quad \rho v \quad = -\nabla\cdot(\rho\vec{u}\vec{u}^T + pI)$$

$$dW = d\vec{F}_p\cdot\vec{u} = -p\,\vec{n}\cdot\vec{u}\,ds$$

↓ scalar

$$\frac{d}{dt}\int E = -\int_{\partial\Omega} E\,\vec{u}\,\vec{n} - \int_{\partial\Omega} p\,\vec{u}\cdot\vec{n}\,ds$$

$$\frac{d}{dt}\int E = -\int_{\partial\Omega}(E+p)\,\vec{u}\cdot\vec{n}\,ds$$

$$E_t = -\nabla\cdot(E+p)\,\vec{u}$$

Conservative $U = \begin{bmatrix} U_1 \\ \vdots \\ U_4 \end{bmatrix} = \begin{bmatrix} \rho \\ m_x \\ m_y \\ E \end{bmatrix}$ → density; $m_x = \rho u$, $m_y = \rho v$ momentum's density; → energy

(2.3) (2.4)

$$e = \frac{E}{\rho} - \frac{1}{2}|u|^2 \qquad p = (\gamma-1)\rho e = (\gamma-1)\left(E - \frac{m_x^2+m_y^2}{2\rho}\right) \quad (2.4)$$

$$E = \frac{p}{\gamma-1} + \frac{1}{2}\rho|u|^2$$

Given γ

prim: $\begin{pmatrix} \rho \\ u \\ v \\ p \end{pmatrix}$ → Conserve $\begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \frac{1}{2}(\rho u^2+\rho v^2) + \frac{p}{\gamma-1} \end{pmatrix}$

Flux → $\vec{F} = \begin{pmatrix} \rho \vec{u}^T \\ \rho u \otimes u + pI \\ (E+p)\vec{u}^T \end{pmatrix}$



$\vec{F}_y$ flux_x_vec

function $F(U) = \begin{pmatrix} \rho u & \rho v \\ \rho u u & \rho u v \\ \rho u v & \rho v v \\ E u & E v \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ p & 0 \\ 0 & p \\ pu & pv \end{pmatrix}$

1x 1y

each row

$U_t = -\nabla \cdot F(U)$

$\vec{F}_x$ ↳ Flux in x-direction not derivative!

$U_t + \nabla \cdot F(u) = 0$

$U_t + \partial_x F_{\vec{x}} + \partial_y F_{\vec{y}} = 0$

Roe average: Refer H.C. Yee 1989

The conservation laws for the one-dimensional Euler equations for equilibrium real gases can be written in the form

$$\frac{\partial U}{\partial t} + \frac{\partial F(U)}{\partial x} = 0, \tag{4.1a}$$

where the column vectors U and $F(U)$ take the form

$$U = \begin{bmatrix} \rho \\ m \\ \epsilon \end{bmatrix}, \qquad F = \begin{bmatrix} \rho u \\ mu + p \\ \epsilon u + pu \end{bmatrix}. \tag{4.1b}$$

Here ρ is the density, $m = \rho u$ is the momentum per unit volume, p is the pressure, $\epsilon = \rho(\epsilon + \frac{1}{2}u^2)$ is the total internal energy per unit volume, and ϵ is the internal energy per unit mass. Note that there is an arbitrary additive constant in the definition of ϵ.

To write as our notation $E = \rho(e + \frac{1}{2}|u|^2)$ different with Shu's notation

$$h = e + \frac{p}{\rho}$$

where the thermodynamic derivatives are defined as

$$\chi = \left(\frac{\partial p}{\partial \rho}\right)_{\tilde{\epsilon}}; \qquad \kappa = \left(\frac{\partial p}{\partial \tilde{\epsilon}}\right)_{\rho}. \tag{4.2b}$$

Here the subscript $\tilde{\epsilon}$ (ρ) means the partial derivative of p with respect to ρ ($\tilde{\epsilon}$) by holding $\tilde{\epsilon}$ (ρ) constant. If $h = \epsilon + p/\rho$ is the specific enthalpy, one can obtain for the speed of sound c the relation

$$c^2 = \chi + \kappa h. \tag{4.3}$$

For a perfect gas with $p = \rho\epsilon(\gamma - 1)$, where γ is the ratio of specific heat, $\chi = 0$, and $\kappa = (\gamma - 1)$. For a thermally perfect gas $p = \rho w(\epsilon)$, the temperature $T = w(\epsilon)$ where w is a function of ϵ only, and $\chi = w(\epsilon)$.

The Jacobian matrix A takes the form

$$A = \begin{bmatrix} 0 & 1 & 0 \\ \chi - (2 - \kappa)u^2/2 & (2 - \kappa)u & \kappa \\ (\chi + \kappa u^2/2 - H)u & H - \kappa u^2 & (1 + \kappa)u \end{bmatrix}, \tag{4.4}$$

where $H = h + u^2/2$ is the total enthalpy. The three eigenvalues of A are

$$a^1 = u - c, \quad a^2 = u, \quad \text{and} \quad a^3 = u + c. \tag{4.5}$$

$$H = e + \frac{p}{\rho} + \frac{|u|^2}{2} = \frac{p}{(\gamma-1)\rho} + \frac{p}{\rho} + \frac{|u|^2}{2} = \frac{1}{\rho}\left(\frac{p}{\gamma-1} + \frac{1}{2}|u|^2\rho + p\right)$$

$$= \frac{1}{\rho}(E + p)$$

gas, the Roe averaged state can readily be obtained as

$$\overline{u} = \frac{u^L + Du^R}{1 + D}, \tag{4.11a}$$

$$\overline{H} = \frac{H^L + DH^R}{1 + D}, \tag{4.11b}$$

$$\overline{c}^2 = (\gamma - 1)\left[\overline{H} - \frac{1}{2}\overline{u}^2\right], \tag{4.11c}$$

with

$$D = \sqrt{\frac{\rho^R}{\rho^L}}. \tag{4.11d}$$

Using this in Riemann Solver

Riemann Solver: Harten_lax_van_Leer Ship 8 (2.11)

HLLC Since for a given cell, 

refer [7]
Also didn't answer how to choose γ.

We note inner ↑ outer

Input (cons_in | cons_out
gamma_in | gamma_out)

inner ⟶ outer

normal vector ↓

$P_{in}\ u_{in}\ v_{in}\ E_{in} \mid P_{out}\ u_{out}\ v_{out}\ E_{out}$

$u_{n_in} = \begin{pmatrix} u_{in} \\ v_{in} \end{pmatrix} * \vec{n}$ $u_{n_out} = \begin{pmatrix} u_{out} \\ v_{out} \end{pmatrix} * \vec{n}$ $c_{in} = \sqrt{\frac{\gamma_{in} P_{in}}{\rho_{in}}}$ $c_{out} = \sqrt{\frac{\gamma_{out} P_{out}}{\rho_{out}}}$

$u_{t_in} = \vec{u}_{in} \times \vec{t}$ $u_{t_out} = u_{out} * \vec{t}$

$D = \sqrt{\frac{P_{out}}{P_{in}}}$

$u_{n_Roe} = \frac{u_{n_in} + D \cdot u_{n_out}}{1+D}$

$u_{t_Roe} = \frac{u_{t_in} + D \cdot u_{t_out}}{1+D}$

$H_{in} = \frac{E_{in} + P_{in}}{\rho_{in}} \Rightarrow H_{Roe} = \frac{H_{in} + D H_{out}}{1+D}$

$H_{out} = \frac{E_{out} + P_{out}}{\rho_{in}}$

$$c_{Roe} = \sqrt{(\gamma - 1)\left(H_{Roe} - \frac{1}{2}|u_{Roe}|^2\right)}$$

$\gamma = \frac{1}{2}(\gamma_{in} + \gamma_{out})$

This is a huge bug since I have nothing to refer, I just do one ...

$$m_{u_min} = \min\{u_{n_in} - c_{in},\ u_{n_Roe} - c_{Roe}\}$$

$$m_{u_plus} = \max\{u_{n_out} + c_{out},\ u_{n_Roe} + c_{Roe}\}$$

Numerator_mu_Star =

$$\rho_{out}\, u_{n_out}(mu_{plus} - u_{n_out}) - \rho_{in}\cdot u_{n_in}(mu_{minus} - u_{n_in}) + P_{in} - P_{out}$$

Denominator_mu_Star =

$$\rho_{out}(m u_{plus} - u_{n_out}) - \rho_{in}(mu_{minus} - u_{n_in})$$

$$\checkmark\ mu_Star = \frac{\text{Numerator_mu_Star}}{\text{Denominator_mu_Star}}$$

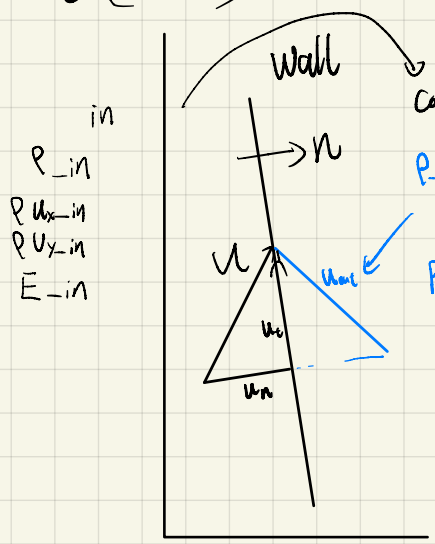
$$\checkmark\ P_{Star} = \rho_{in}\cdot(u_{n_in} - u_{minus})(u_{n_in} - mu_Star) + P_{in}$$

$$\text{Flux}: \vec{G}\cdot n = \begin{bmatrix} 0 \\ P_{Star}\cdot \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} \\ P_{Star} * mu_Star \end{bmatrix}$$

Boundary condition in Euler

reflect ( )

$u_n = (\rho u_x, \rho u_y) \cdot \vec{n}$ $u_t = (\rho u_x, \rho u_y) \cdot \vec{t}$



in

ρ_in

ρu_{x_in}

ρu_{y_in}

E_in

Construct

$\rho_out = \rho_in$ $u_{out} = \vec{u_t} - \vec{u_n}$

$\rho_out = \rho_in$ $(\vec{u} = \vec{u_t} + \vec{u_n})$

$= \vec{u} - 2\vec{u_n}$

$= \vec{u} - 2(\vec{u} \cdot \vec{n})\vec{n}$

Return ρ_out

E_out

function left_eigenvec_Jacobian

In this work, we use the local characteristic decomposition for the jumps of the damping coefficients. Define $\mathbf{R}^{-1}$ as a 4×4 matrix with the left eigenvectors of Jacobian matrix $\mathbf{F}(\overline{\mathbf{U}}) \cdot \mathbf{n}_l$, where $\mathbf{n}_l = (n_x, n_y)^T$, $l = 1, 2, 3, 4$ is the outward unit normal to different edges of the cell I_{ij}, and $\overline{\mathbf{U}}$ is the cell average vector of the conserved variables. In the two-dimensional compressible Euler equations of the perfect gas, $\mathbf{R}^{-1}$ has the following

expression

set

$$\mathbf{R}^{-1} = \frac{\gamma - 1}{c} \begin{pmatrix} \frac{1}{2}(B_r + \tilde{u}c) & -\frac{1}{2}((\gamma-1)u + n_x c) & -\frac{1}{2}\left((\gamma-1)v + n_y c\right) & \frac{1}{2}(\gamma-1) \\ \tilde{v}c & n_y c & -n_x c & 0 \\ c^2 - B_r & (\gamma-1)u & (\gamma-1)v & -(\gamma-1) \\ \frac{1}{2}(B_r - \tilde{u}c) & -\frac{1}{2}((\gamma-1)u - n_x c) & -\frac{1}{2}\left((\gamma-1)v - n_y c\right) & \frac{1}{2}(\gamma-1) \end{pmatrix}, \tag{3.4}$$

self

where $\tilde{u} = un_x + vn_y$, $\tilde{v} = -un_y + vn_x$, and $B_r = (\gamma - 1)\frac{u^2+v^2}{2}$. Then the vector $\partial^\alpha \boldsymbol{V}|_{\boldsymbol{v}} = (\partial^\alpha V_1, \ldots, \partial^\alpha V_4)^T\Big|_{\boldsymbol{v}}$ is obtained by $\partial^\alpha \boldsymbol{V}|_{\boldsymbol{v}} = \mathbf{R}^{-1}\big|_{\mathbf{n}_l} \partial^\alpha \mathbf{U}|_{\boldsymbol{v}}$.

for each of them they get it!

Done